



INTERNSHIP REPORT ON
“ FULL STACK DEVELOPMENT WITH MERN
VIRTUAL INTERNSHIP”
EDUNET FOUNDATION

Submitted by,

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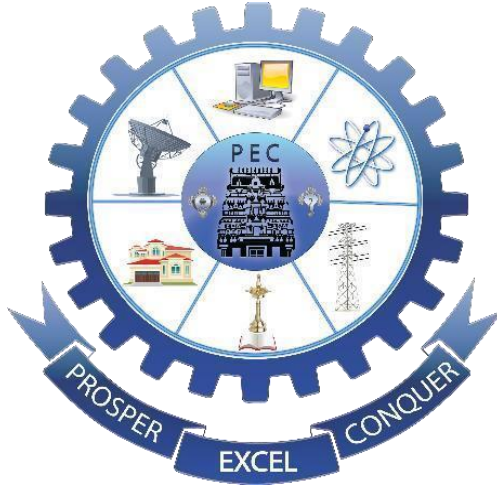
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COLLEGE: PAAVAI ENGINEERING COLLEGE

PLACE: PACHAL, NAMAKKAL

PIN CODE: 637018



UNDER THE GUIDANCE OF
EDUNET FOUNDATION

DECLARATION

I hereby declare that the internship training entitled “**FULL STACK DEVELOPMENT WITH MERN VIRTUAL INTERNSHIP**” Submitted to the department of **INFORMATION TECHNOLOGY** in **PAAVAI ENGINEERING COLLEGE, NAMAKKAL** for the industrial internship in the degree of **BACHELOR OF TECHNOLOGY** is a record of original work done by me under the guidance of **EDUNET FOUNDATION**.

PLACE: NAMAKKAL

DATE:

SIGNATURE OF HOD

SIGNATURE OF STUDENT

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ABSTRACT

The **Building Modern Web Applications with MERN Stack Internship**, organized by **EY Global Delivery Services in collaboration with AICTE and Edunet Foundation under the Next Gen Employability Program**, provided an industry-oriented platform to enhance knowledge and practical skills in full-stack web development. The internship was structured around guided self-learning modules and hands-on projects that covered both the fundamental and advanced concepts of modern web application development using the MERN technology stack.

Through this internship, I gained proficiency in core technologies such as **MongoDB, Express.js, React.js, and Node.js**, which together form a powerful framework for developing scalable and dynamic web applications. The program provided practical exposure to frontend development using React components, backend API creation using Node.js and Express.js, and database management using MongoDB. Additionally, the internship covered important concepts such as REST API integration, CRUD operations, responsive UI design, state management, and full-stack application architecture.

The program emphasized **project-based learning**, enabling participants to design, develop, test, and deploy modern web applications aligned with real-world industry practices. During the internship, I worked on various assignments and projects that strengthened my understanding of frontend-backend integration, data handling, debugging techniques, and software development workflows. These activities helped improve my analytical thinking, coding skills, and ability to build interactive and user-friendly applications.

In today's rapidly evolving technology landscape, practical experience in full-stack development is essential for aspiring software engineers. The MERN Stack Internship program provided an immersive and structured learning pathway that helped students gain hands-on experience in modern web technologies. The program included structured learning modules, project implementation, and real-time coding exercises that enhanced both conceptual understanding and practical skills

LEARNING OBJECTIVES / INTERNSHIP OBJECTIVES

The primary objectives of the **Building Modern Web Applications with MERN Stack Internship** are as follows:

1. **Understand Full-Stack Web Development Fundamentals** – Gain a strong foundation in modern web development architecture including frontend, backend, and database integration using the MERN stack.
2. **Learn Modern Web Development Tools & Frameworks** – Acquire hands-on experience with technologies such as **MongoDB, Express.js, React.js, and Node.js** for building scalable and efficient web applications.
3. **Develop Interactive User Interfaces** – Design and implement responsive and dynamic user interfaces using **React.js components, state management, and modern UI practices**.
4. **Apply Data Handling & Database Management Techniques** – Learn to work with **MongoDB databases**, perform CRUD operations, and manage application data efficiently.
5. **Enhance Problem-Solving Skills** – Strengthen analytical and logical thinking by solving real-time web development challenges and debugging application errors.
6. **Work with Backend APIs & Server Logic** – Explore backend development using **Node.js and Express.js**, including REST API creation and server-side application logic.
7. **Bridge Academic Knowledge with Industry Practices** – Gain exposure to professional coding standards, project development workflows, version control practices, and deployment strategies used in the IT industry.
8. **Improve Collaboration & Professional Growth** – Develop teamwork, communication, project management, and time management skills essential for success in software development careers.

INTRODUCTION TO THE INTERNSHIP PROGRAM

The **Building Modern Web Applications with MERN Stack Internship**, organized by **EY Global Delivery Services in collaboration with AICTE and Edunet Foundation under the Next Gen Employability Program**, provided students with practical exposure to modern web application development. The program focused on full-stack development concepts including frontend design using React.js, backend development using Node.js and Express.js, database management using MongoDB, and project-based learning. It served as a platform to bridge academic knowledge with industry requirements while enhancing technical and problem-solving skills.

1. Introduction:

- The MERN Stack Internship is an industry-oriented initiative designed to provide students with hands-on exposure to modern web application development.
- It bridges the gap between theoretical knowledge gained in academics and the practical skills required by the IT industry.
- The program is delivered through online learning sessions, making it accessible to students across India through structured learning modules, assignments, and project work.
- It emphasizes self-paced learning, real-time coding practice, project-based assessments, and mentorship to encourage independent learning and innovation.

2. About AICTE & Edunet Foundation:

- AICTE (All India Council for Technical Education) is a statutory body and national-level council for technical education under the Department of Higher Education, Ministry of Education, Government of India.
- It focuses on improving the quality of technical education and enhancing student employability through skill-based training programs and internships.
- Edunet Foundation is a non-profit organization working with academia, government, and industry to create a strong skill-based ecosystem in emerging technologies.
- Edunet collaborates with global technology companies and educational institutions to provide students with opportunities to develop industry-ready skills.
- Together, AICTE and Edunet Foundation bring internship opportunities to students across India, helping them gain practical experience and certifications recognized by leading organizations.

3. About EY Global Delivery Services Program:

- EY Global Delivery Services is part of Ernst & Young (EY), one of the world's leading professional services organizations providing consulting, assurance, tax, and advisory services.
- The MERN Stack internship program is designed to equip students with the skills required to build modern, scalable, and efficient web applications.
- It includes structured modules covering frontend development with React.js, backend development with Node.js and Express.js, database management using MongoDB, and full-stack integration.
- The program provides access to learning resources, practical assignments, and real-time project development practices aligned with industry standards.
- Successful completion of the program helps students gain certification from EY, AICTE, and Edunet Foundation, enhancing their employability and readiness for careers in web development.

COMPANY / PLATFORM PROFILE

Edunet Foundation is an organization dedicated to developing digital and technical skills among students and young professionals. It focuses on bridging the gap between academic learning and industry requirements by providing training programs, internships, and certification courses in emerging technologies.

Vision:

- To empower students with future-ready digital and technical skills.
- To bridge the gap between academia and industry by providing globally recognized learning opportunities.
- To build a strong ecosystem of skilled professionals contributing to digital transformation and technological growth.

Mission:

- To make world-class technology education accessible to students through virtual internships and structured training programs.
- To collaborate with global industry leaders such as EY and technology organizations to deliver industry-relevant training.

- To enhance employability, entrepreneurship, and technical skills among students across the country.

Business / Program Overview:

- AICTE and Edunet Foundation partnered with EY Global Delivery Services to deliver the MERN Stack internship program.
- The program offers structured modules covering React.js frontend development, Node.js backend programming, Express.js frameworks, and MongoDB database management.
- It is designed as a self-paced, project-based internship that includes real-time coding exercises, assignments, and project implementation.
- Students completing the program receive industry-recognized certification from EY, AICTE, and Edunet Foundation, which adds strong value to their professional career.

Key Services:

- Virtual internship opportunities in emerging technologies such as Web Development, Artificial Intelligence, Cloud Computing, Cybersecurity, and Data Science.
- Online learning platforms with structured modules, recorded lectures, and hands-on assignments.
- Industry-recognized certifications jointly offered with leading technology partners.
- Mentorship, career guidance, and skill development programs for students.
- Bridging the gap between academic education and industry requirements through practical and project-based learning.

INTERNSHIP STRUCTURE & MODULES

The **Building Modern Web Applications with MERN Stack Internship** was designed in a self-paced, module-based structure, allowing students to progressively build their knowledge and skills in modern web application development. The internship combined theoretical learning with practical implementation, ensuring that participants could apply concepts in real-world projects. The program focused on full-stack development using the MERN technology stack and provided hands-on experience through assignments, coding exercises, and project development.

The program was divided into two major learning tracks, each containing multiple modules:

Course 1: Frontend Development with React.js/html

- Introduction to modern web development and development environment setup
- Understanding JavaScript fundamentals and ES6 features
- Building user interfaces using React.js components and JSX
- Managing application state and props in React applications
- Designing responsive user interfaces and handling user interactions

Course 2: Backend Development with Node.js and MongoDB

- Introduction to Node.js and Express.js for backend application development
- Creating RESTful APIs and handling HTTP requests and responses
- Database integration using MongoDB and Mongoose
- Implementing CRUD operations for managing application data
- Connecting frontend applications with backend APIs for full-stack functionality
- Application testing, debugging, and preparing web applications for deployment

WORK ASSIGNMENT / TASKS

WEEK 1 → [10/02/2025 to 16/02/2025] (Introduction to MERN Stack & Development Setup):

1. Installation and setup of Node.js, MongoDB, and development tools.
2. Introduction to MERN stack architecture (MongoDB, Express.js, React.js, Node.js).
3. Understanding JavaScript ES6 concepts and project structure for web applications.

WEEK 2 → [17/02/2025 to 23/02/2025] (Frontend Development – React.js):

1. Learning React.js fundamentals, components, and JSX syntax.
2. Designing user interfaces using React components and CSS frameworks.
3. Creating a simple React-based user interface application.

WEEK 3 → [24/02/2025 to 02/03/2025] (Backend Development – Node.js & Express.js):

1. Introduction to Node.js server environment and Express.js framework.

2. Creating RESTful APIs for handling client requests.
3. Developing a simple backend API for managing application data.

WEEK 4 → [03/03/2025 to 09/03/2025] (Database Integration – MongoDB):

1. Introduction to MongoDB database and NoSQL data modeling.
2. Implementing CRUD operations using MongoDB and Mongoose.
3. Integrating database operations with backend APIs.

WEEK 5 → [10/03/2025 to 16/03/2025] (Full Stack Integration):

1. Connecting React frontend with Node.js backend APIs.
2. Implementing data exchange between frontend and MongoDB database.
3. Mini Project – Developing a basic full-stack MERN web application.

WEEK 6 → [17/03/2025 to 21/03/2025] (Final Project Development & Testing):

1. Development of the final full-stack web application using MERN stack.
2. Testing and debugging the application to ensure smooth functionality.
3. Capturing screenshots of the project and preparing documentation for the internship report.
4. Submission of the final project and completion of internship evaluation.

About MERN Stack & Development Tools:

- The **MERN Stack** is a popular technology stack used for developing modern web applications. It consists of **MongoDB, Express.js, React.js, and Node.js**, which together enable developers to build full-stack applications using JavaScript.
- It provides a complete environment for developing dynamic, scalable, and high-performance web applications.
- The stack supports both **frontend and backend development**, making it easier to manage the entire application using a single programming language.
- Modern development tools such as **Visual Studio Code, Node Package Manager (NPM)**,

and Git are commonly used along with MERN stack technologies.

- MERN stack applications allow developers to build responsive user interfaces, manage server-side logic, and store application data efficiently.

How MERN Stack Works?

1. Development Environment

- The MERN stack development environment typically uses tools such as **Visual Studio Code**, which provides a powerful code editor for writing and managing application code.
- Developers write frontend code using **React.js**, backend code using **Node.js and Express.js**, and manage databases using **MongoDB**.

2. Project Structure

- A MERN stack project usually consists of two main parts: the **frontend (React application)** and the **backend server (Node.js and Express.js)**.
- The backend communicates with the **MongoDB database** to store and retrieve application data, while the frontend handles user interactions and interface design.
- This structured approach ensures applications are **scalable, modular, and easy to maintain**.

3. Code Editing & Development Tools

- Developers use code editors such as **Visual Studio Code** with extensions that support JavaScript, React, and Node.js development.
- The editor provides features such as **syntax highlighting, auto-completion, debugging tools, and integrated terminal**, which improve development productivity.

4. Package Management & Dependencies

- The MERN stack uses **Node Package Manager (NPM)** or **Yarn** to install and manage external libraries and dependencies required for application development.
- These tools help developers easily integrate frameworks, middleware, and utilities needed for building web applications.

5. Testing & Debugging

- Debugging tools such as **browser developer tools, console logs, and Node.js debugging utilities** help developers monitor application behavior and identify errors.
- Testing ensures the web application runs smoothly and handles user requests efficiently.

6. Deployment & Application Hosting

- After development and testing, MERN stack applications can be deployed on platforms such as **Vercel, Netlify, Heroku, or cloud services**.
- This allows users to access the application through the internet using web browsers.

Services:

1. **Virtual Internships** – AICTE and Edunet Foundation provide online internship opportunities in collaboration with organizations such as **EY Global Delivery Services**.
2. **Skill Development Programs** – Structured training programs in emerging technologies including **Web Development, Artificial Intelligence, Cloud Computing, Cybersecurity, and Data Science**.
3. **Industry-Recognized Certifications** – Students receive certificates jointly recognized by **AICTE, Edunet Foundation, and industry partners**.
4. **Learning Management System (LMS)** – Access to online learning platforms containing training modules, recorded sessions, assignments, and assessments.
5. **Project-Based Learning** – Hands-on projects and assignments that allow students to apply concepts and develop real-world applications.
6. **Mentorship & Career Guidance** – Guidance from industry experts and mentors to improve technical knowledge and career readiness.
7. **Collaboration with Academia & Industry** – Strengthening the connection between academic education and industry practices through skill-based training programs.

Development Environment:

The **MERN Stack** is a full-stack development framework used to build modern and scalable web applications. It combines **MongoDB (Database), Express.js (Backend Framework), React.js (Frontend Library), and Node.js (Runtime Environment)** to create dynamic and responsive applications.

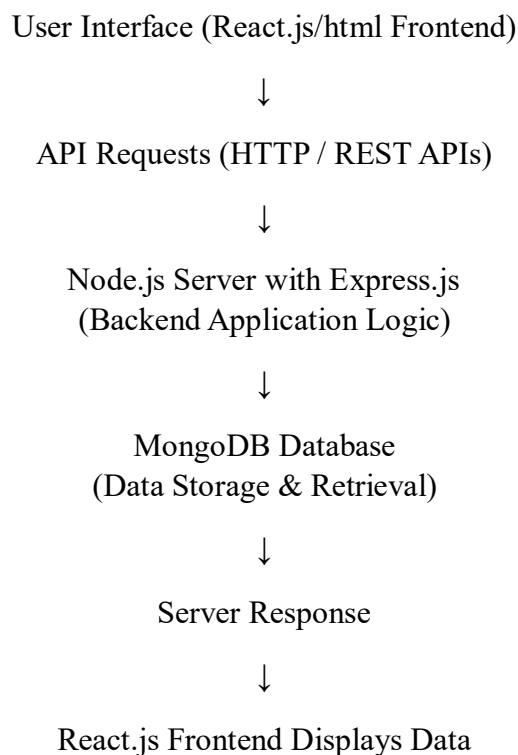
These technologies work together to handle different parts of the application including user interface design, server-side logic, database management, and data communication between components. The MERN stack allows developers to use **JavaScript throughout the entire**

application, which improves development efficiency and simplifies the software development process.

Key Features of MERN Stack:

- Full-stack JavaScript development (Frontend + Backend)
- Component-based user interface using React.js
- Fast and scalable backend using Node.js and Express.js
- NoSQL database management using MongoDB
- REST API communication between client and server
- Flexible and scalable architecture for modern applications

Block Diagram – Android Studio Workflow:



Project Example –Quick Food Deliver App:

The **Quick Food Delivery Web Application** developed during the internship provides a simple user authentication system that allows users to register and log in to the platform through a user-friendly interface. The application uses **React.js for designing the frontend pages**, integrates **Node.js and Express.js for backend API communication**, and stores user information in a **MongoDB database**. The system demonstrates how user data is collected

from the frontend, processed by the server, and securely stored in the database before displaying the welcome dashboard.

Features:

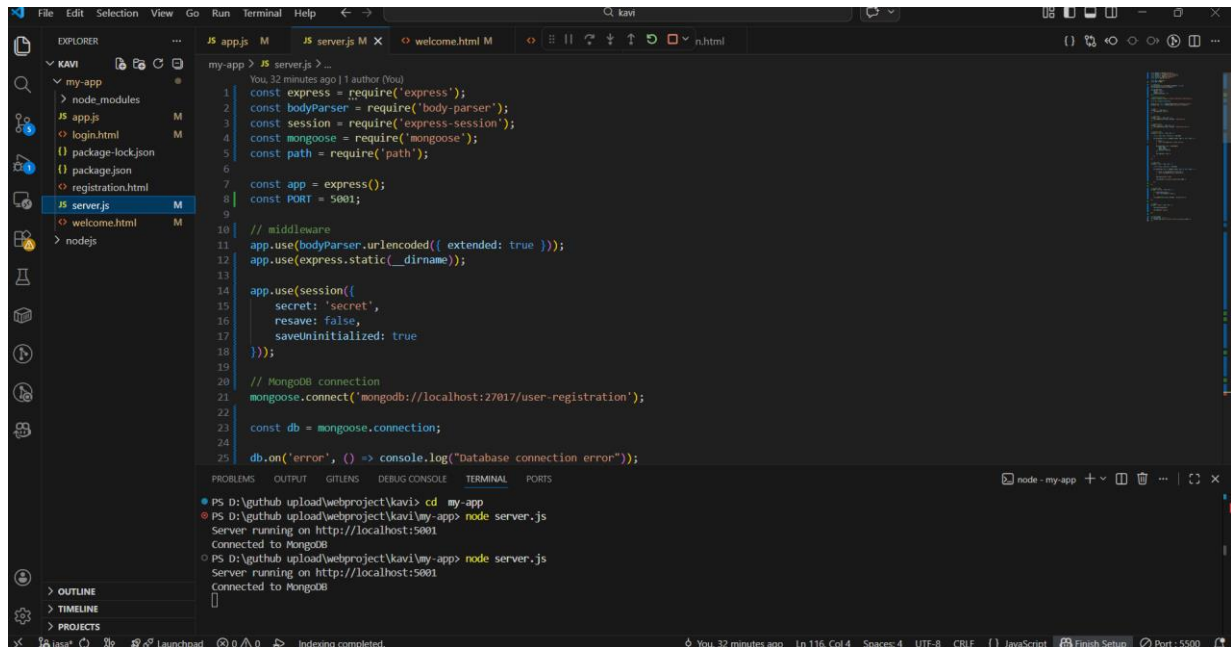
- User can create a new account through the **registration page**.
- Registered users can log in using their **username and password**.
- Displays a **welcome page** after successful login.
- Uses **MongoDB database** for storing user registration details.
- Responsive and simple **user interface built using React.js**.

Project Description:

- **Mini Project 1 – Registration Page:** A simple web page that allows new users to create an account by entering details such as username, email, and password. The entered information is sent to the backend server and stored in the MongoDB database.
- **Mini Project 2 – Login Page:** A login system that enables registered users to access the application using their credentials. The system verifies the user information with the database and allows access if the credentials are valid.
- **Mini Project 3 – Welcome Page:** A basic dashboard page that is displayed after successful login. This page confirms that the user has logged into the system and demonstrates frontend and backend integration.
- **Final Project – Quick Food Delivery Web Application:** A full-stack MERN web application that includes user registration, login authentication, and a welcome interface. The project integrates **React.js for frontend development, Node.js and Express.js for backend APIs, and MongoDB for storing user data**, demonstrating the basic structure of a food delivery platform.

SCREENSHOTS:

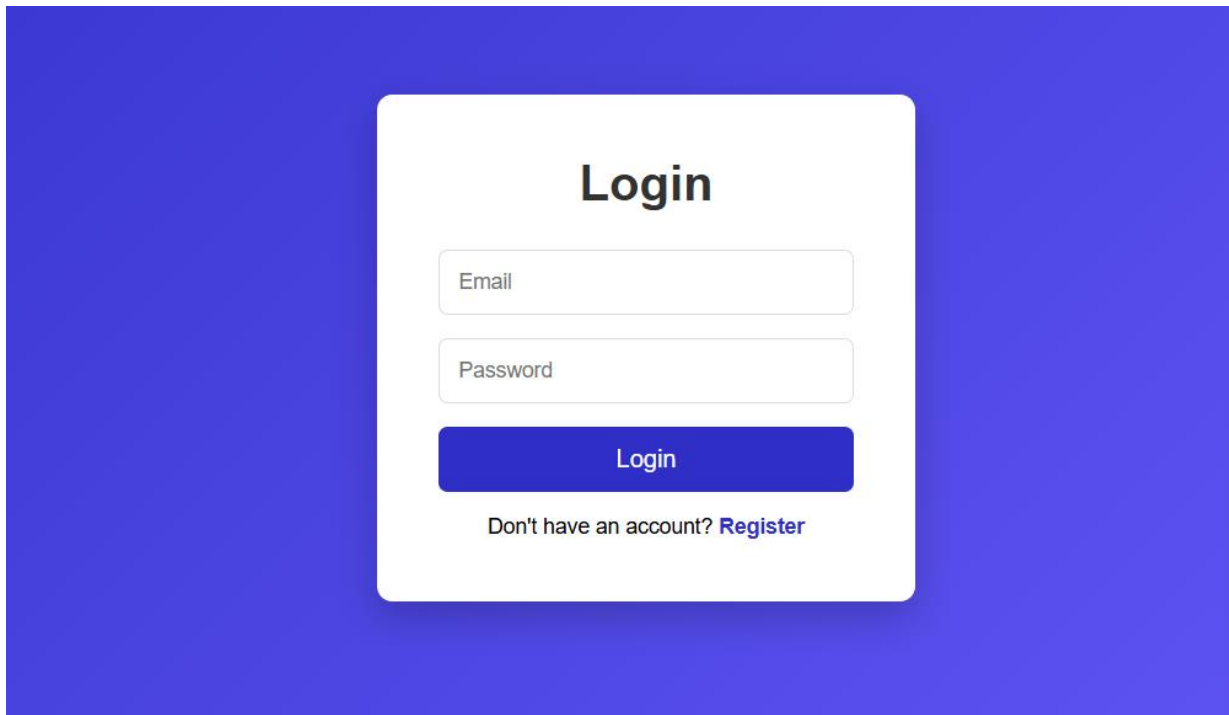
Screenshot 1: MERN Stack project workspace showing the frontend and backend project structure in the development environment.



Screenshot 2: Registration page interface of the Quick Food Delivery web application.

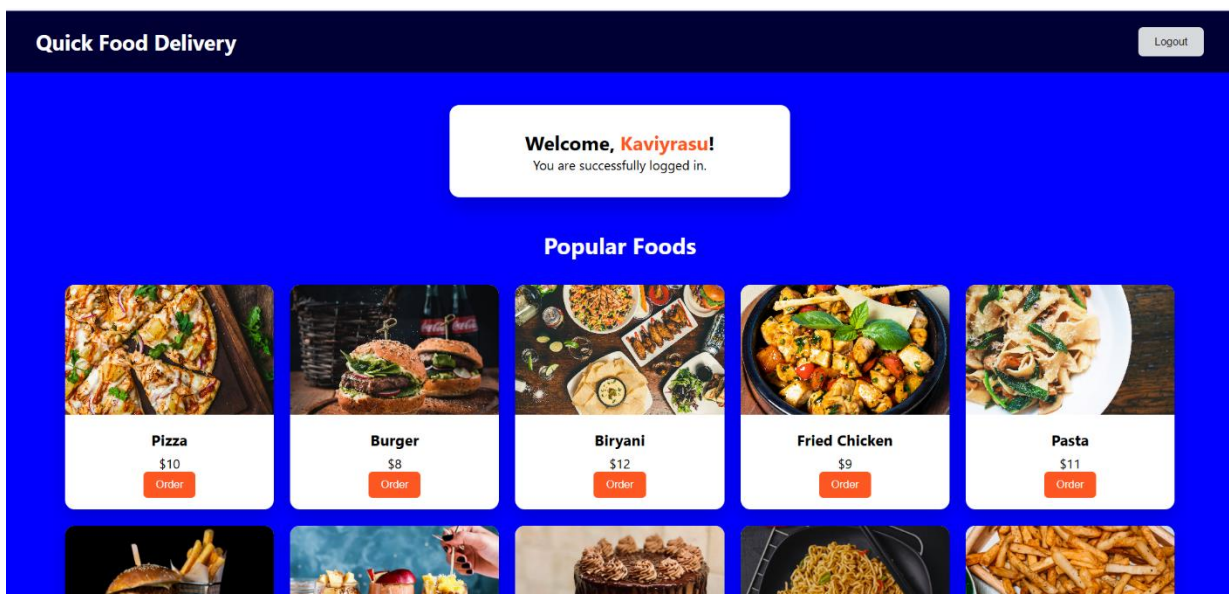
A screenshot of the 'Quick Food Deliver App' registration page. The page has a dark blue header with the app name and a 'Login' button. The main content area is a light blue gradient. In the center, there is a white 'Register' form with the following fields: 'Username:', 'Email:', and 'Password:'. Each field has a corresponding input box. Below the fields is a blue 'Register' button. At the bottom of the form, there is a link: 'If you have already registered, please [login here](#).'

Screenshot 3: Login page where users enter their credentials to access the application



The login page features a solid blue background. In the center is a white rounded rectangle containing the login form. The form has the title "Login" in bold black text. Below the title are two input fields: "Email" and "Password", both with light gray placeholder text. A blue "Login" button is positioned below the password field. At the bottom of the white box, there is a link that says "Don't have an account? [Register](#)".

Screenshot 4: Welcome page displayed after successful login.

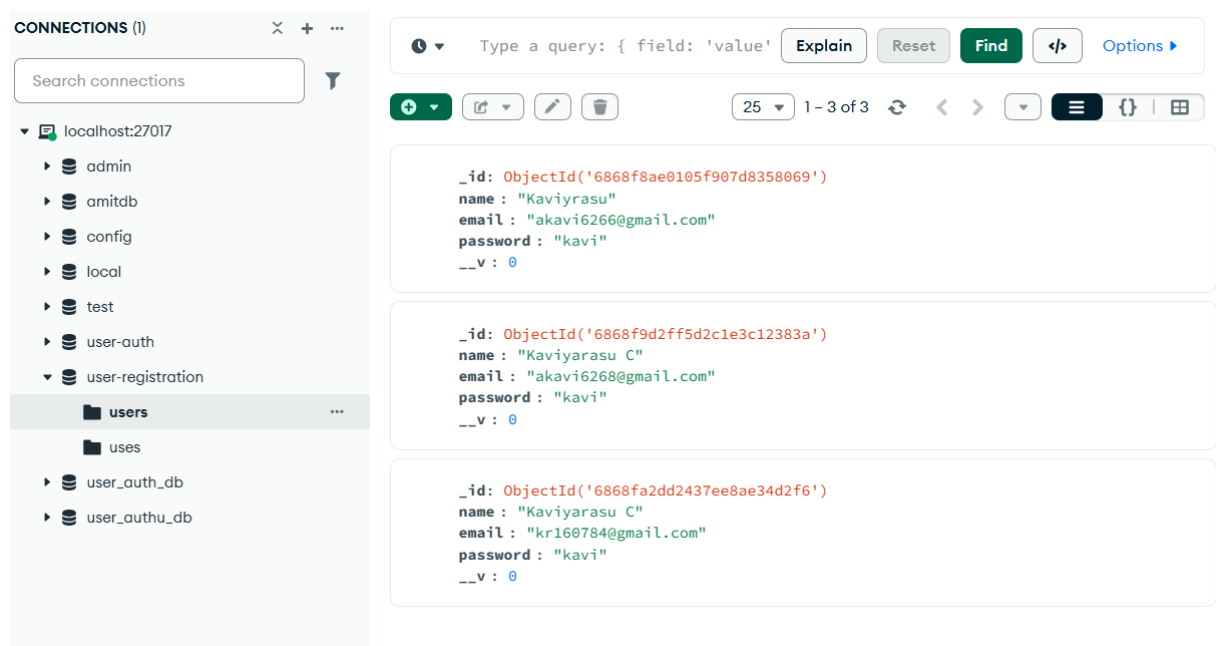


The welcome page has a dark blue header with the text "Quick Food Delivery" on the left and a "Logout" button on the right. Below the header is a white box with the message "Welcome, **Kaviyrasu!**" and "You are successfully logged in." in smaller text. The main section is titled "Popular Foods" and displays a grid of food items. Each item consists of a food image, the item name, its price, and an "Order" button.

Item	Price	Action
Pizza	\$10	Order
Burger	\$8	Order
Biryani	\$12	Order
Fried Chicken	\$9	Order
Pasta	\$11	Order

Below the first row, there are additional food images including a burger with fries, a drink, a cake, a bowl of noodles, and a plate of fries.

Screenshot 5: MongoDB database collection storing registered user information.



MAJOR WORK / PROJECT IMPLEMENTATION

Problem Statement:

- Many beginner web developers struggle to understand how frontend, backend, and database systems work together in modern web applications.
- There is a need for simple, user-friendly web applications that demonstrate core concepts such as **user registration, login authentication, and data storage** in real-world systems like food delivery platforms.
- Lack of practical implementation of **authentication systems and database integration** in beginner projects creates a gap in learning full-stack development skills.

Methodology / Approach:

- Adopted an **Agile development methodology** with incremental development and continuous testing.
- Used **Visual Studio Code** as the primary development environment for building the MERN stack application.
- Implemented the **frontend user interface using React.js** for designing the register, login, and welcome pages.
- Developed **backend APIs using Node.js and Express.js** to handle user authentication and data communication.
- Integrated **MongoDB database** to store and manage user registration details.

- Tested the application using web browsers and debugging tools to ensure proper functionality and user experience.

Implementation:

- **Registration Page:** Developed a page that allows new users to create an account by entering their details. The information is sent to the backend and stored in the MongoDB database.
- **Login Page:** Implemented a login system that verifies user credentials by comparing them with the stored data in the database.
- **Welcome Page:** Created a dashboard page that is displayed after successful login, confirming that the authentication process was completed.
- **Backend APIs:** Built REST APIs using Node.js and Express.js to process user requests and manage database operations.
- **Database Integration:** Stored user details securely in MongoDB using schema-based data models.
- Debugged errors using browser developer tools and performed testing to ensure smooth application functionality.

Results & Outputs:

- Successfully developed a working **Quick Food Delivery Web Application authentication system** including register, login, and welcome pages.
- The application allowed users to **create accounts and securely log into the system**.
- User information was successfully **stored and retrieved from the MongoDB database**.
- The web application displayed a **welcome dashboard after successful login authentication**.
- Gained hands-on experience in **frontend development, backend API creation, and database integration** using the MERN stack.

CHALLENGES & LEARNINGS

Challenges:

- Initial setup issues while installing and configuring **Node.js, MongoDB, and development tools**.
- Understanding the integration between **React frontend and Node.js backend APIs**.
- Debugging errors related to **server connections and database operations**.
- Handling user authentication and ensuring secure data storage in the database.
- Managing project development within the limited internship duration.

Learnings:

- Gained practical knowledge of **MERN Stack technologies including MongoDB, Express.js, React.js, and Node.js.**
- Learned how to design responsive web pages using **React.js components.**
- Understood how **REST APIs communicate between frontend and backend applications.**
- Acquired experience in **database operations such as storing and retrieving user information using MongoDB.**
- Improved problem-solving and debugging skills while developing the project.
- Developed **self-learning, teamwork, and time management skills** during the internship program.

CONCLUSION & FUTURE SCOPE

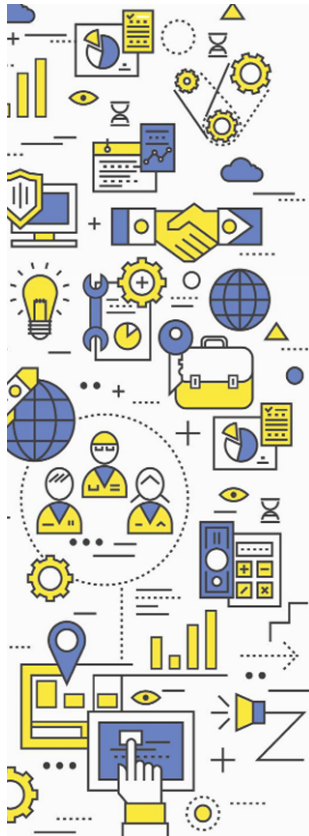
Conclusion:

- The internship provided hands-on exposure to **modern web application development using the MERN Stack.**
- Gained practical skills in **frontend development, backend API development, database integration, and debugging techniques.**
- Successfully developed a **Quick Food Delivery Web Application authentication system** that demonstrates the integration of frontend, backend, and database technologies.
- Improved overall technical knowledge, analytical thinking, and problem-solving abilities through project-based learning.

Future Scope:

- The application can be further enhanced by adding **complete food ordering features such as menu display, cart system, and order management.**
- Future improvements may include **online payment integration and real-time order tracking systems.**
- Integration of **cloud deployment platforms** can make the application accessible to real users.
- Additional security mechanisms such as **JWT authentication and password encryption** can be implemented.
- The skills gained through this internship will serve as a strong foundation for future careers in **full-stack web development and modern application development.**

CERTIFICATE:



CERTIFICATE OF APPRECIATION

Proudly presented to

KAVIYARASU C

STU ID: STU6575608ba8d9d1702191243

for successful completion of the 6 weeks of internship in collaboration with

EY Global Delivery Services & AICTE on
Building Modern Web Applications with MERN Stack
under the **Next Gen Employability Program**
from **February 10, 2025– March 21, 2025.**

Nagesh Singh
Chairman
Edunet Foundation



AICTENG25_43476

Dr. Buddha Chandrasekhar
Chief Coordinating Officer
AICTE

THANK YOU